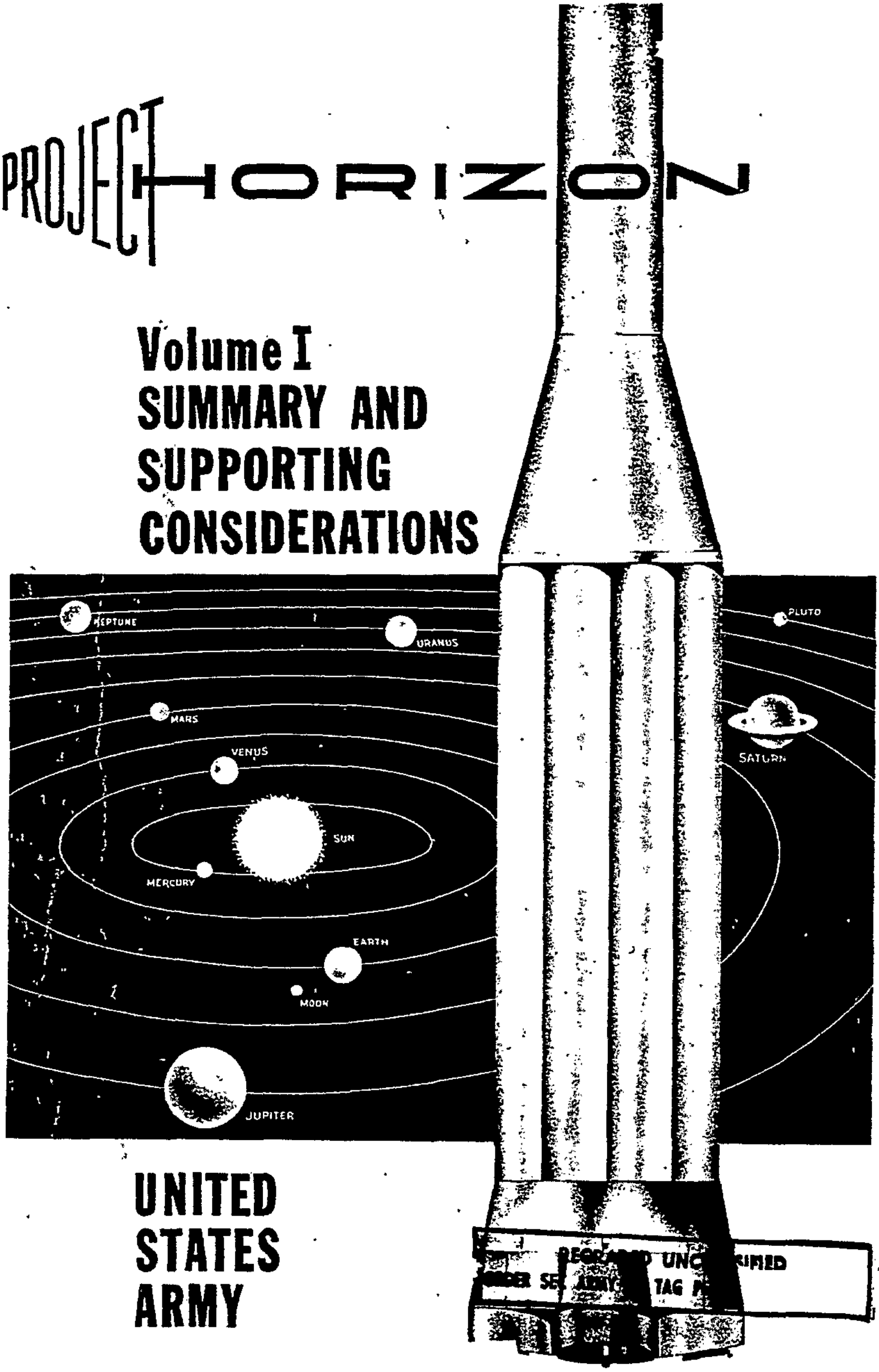
PROJECT HORIZON

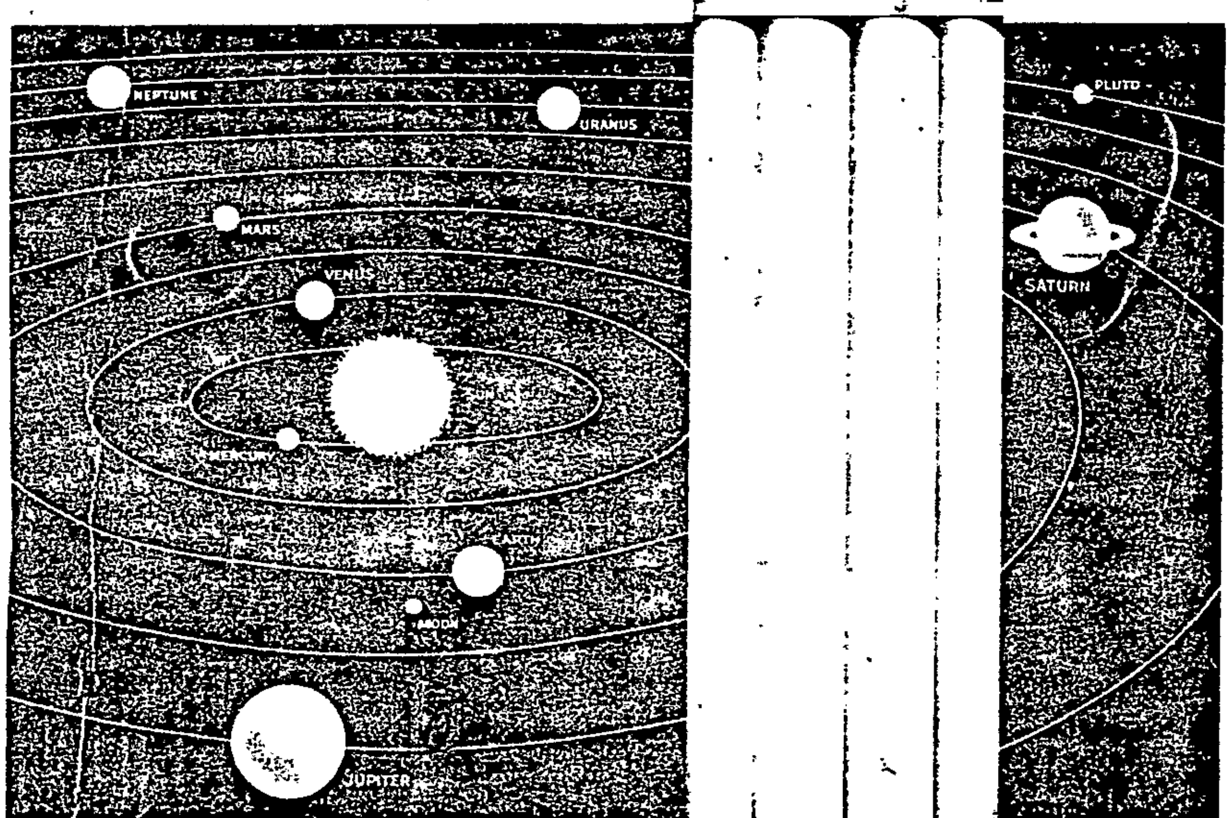
Volume I
SUMMARY AND SUPPORTING CONSIDERATIONS



UNITED
STATES
ARMY

PROJECT HORIZON

Volume II TECHNICAL CONSIDERATIONS & PLANS



UNITED STATES ARMY

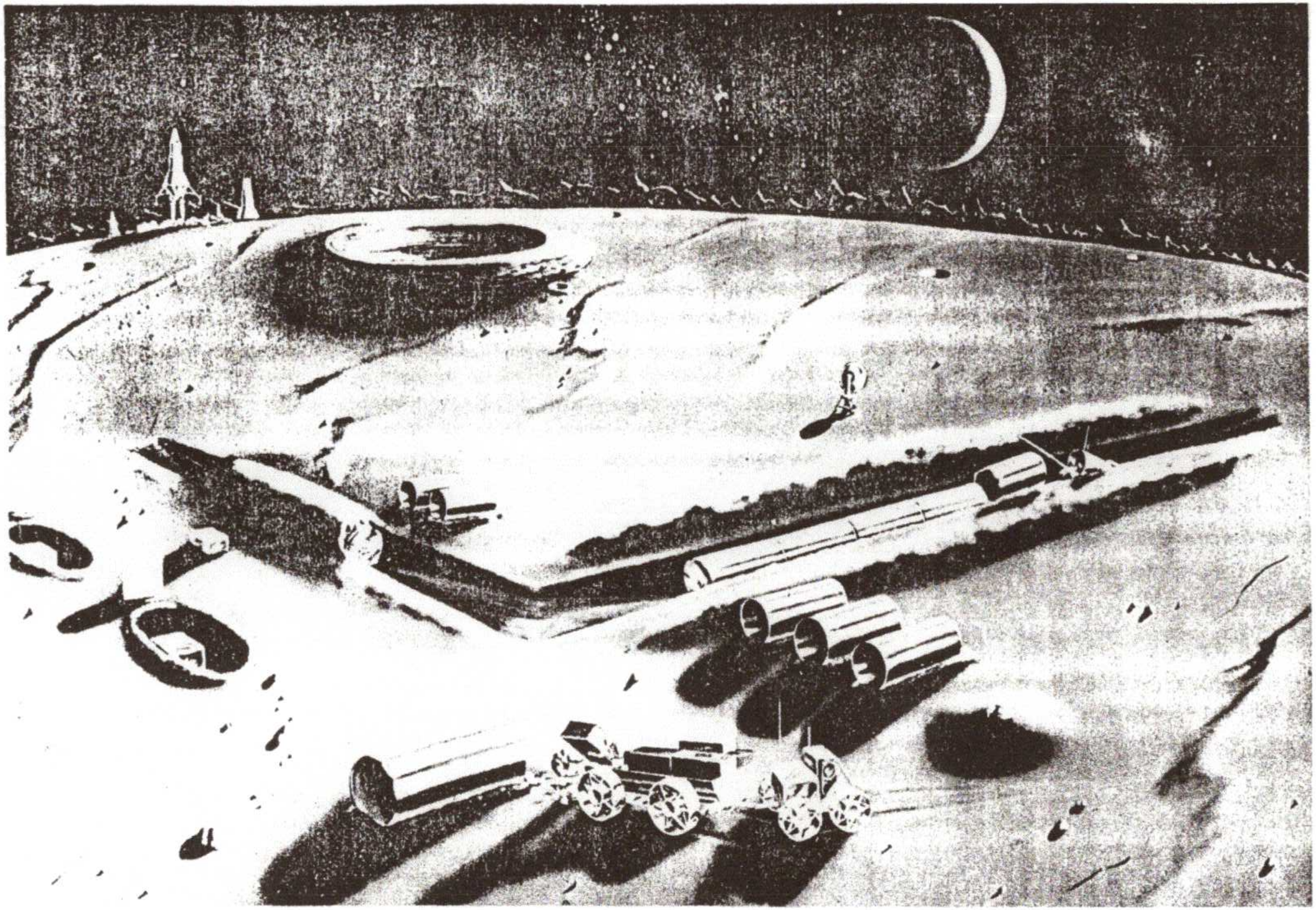


Fig. I-1. HORIZON Outpost in Late 1965

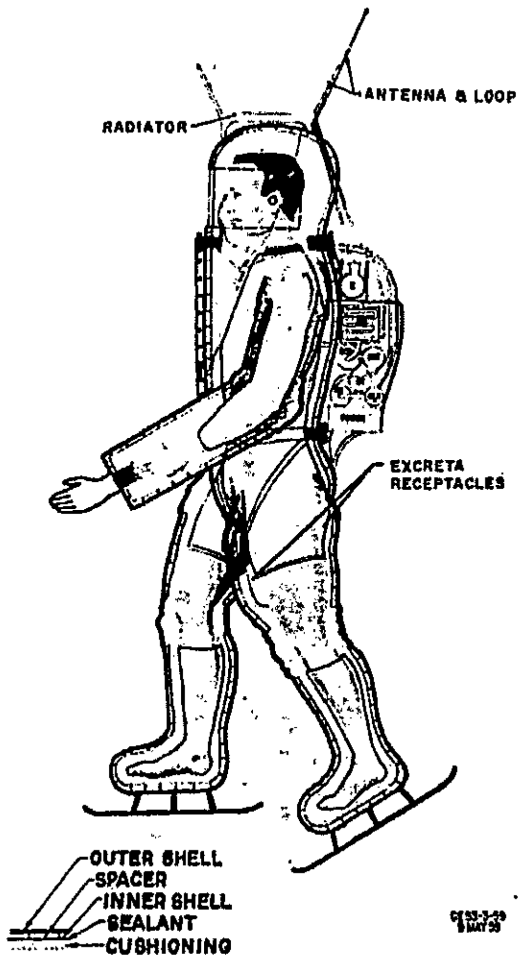


Fig. I-5. Typical Lunar Suit

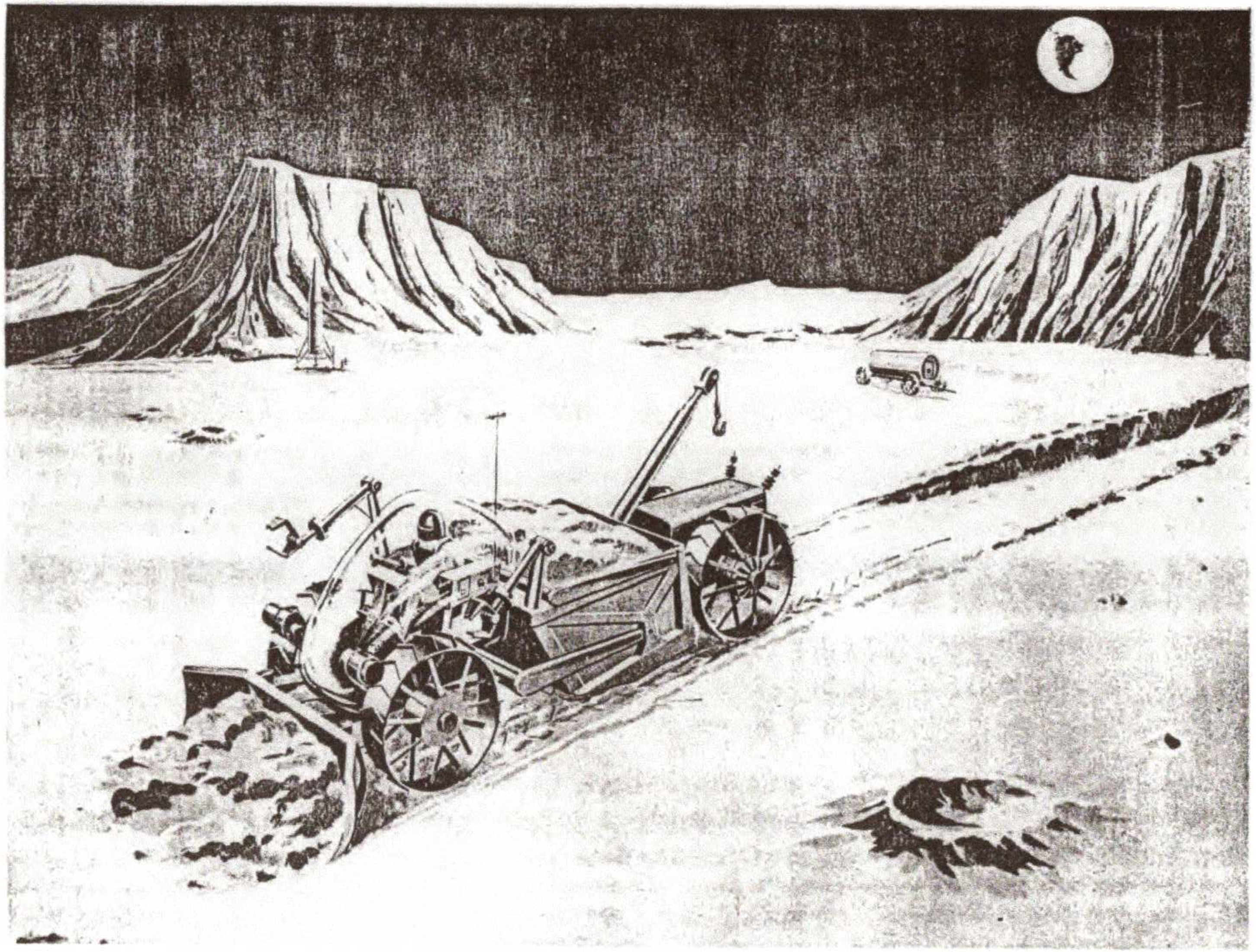


Fig. II-6. Typical Lunar Construction Vehicle

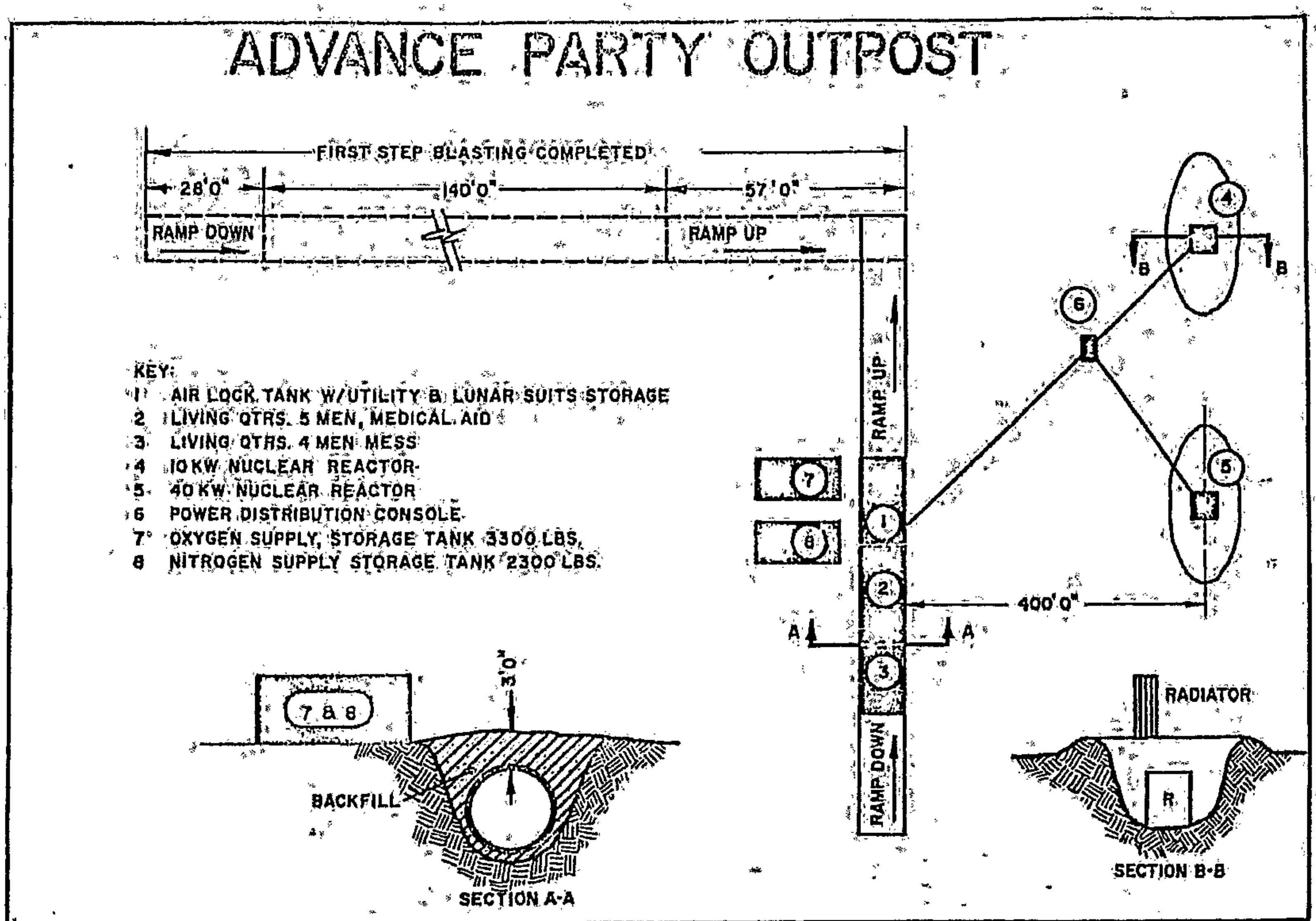


Fig. II-3. Layout of Initial Construction Camp

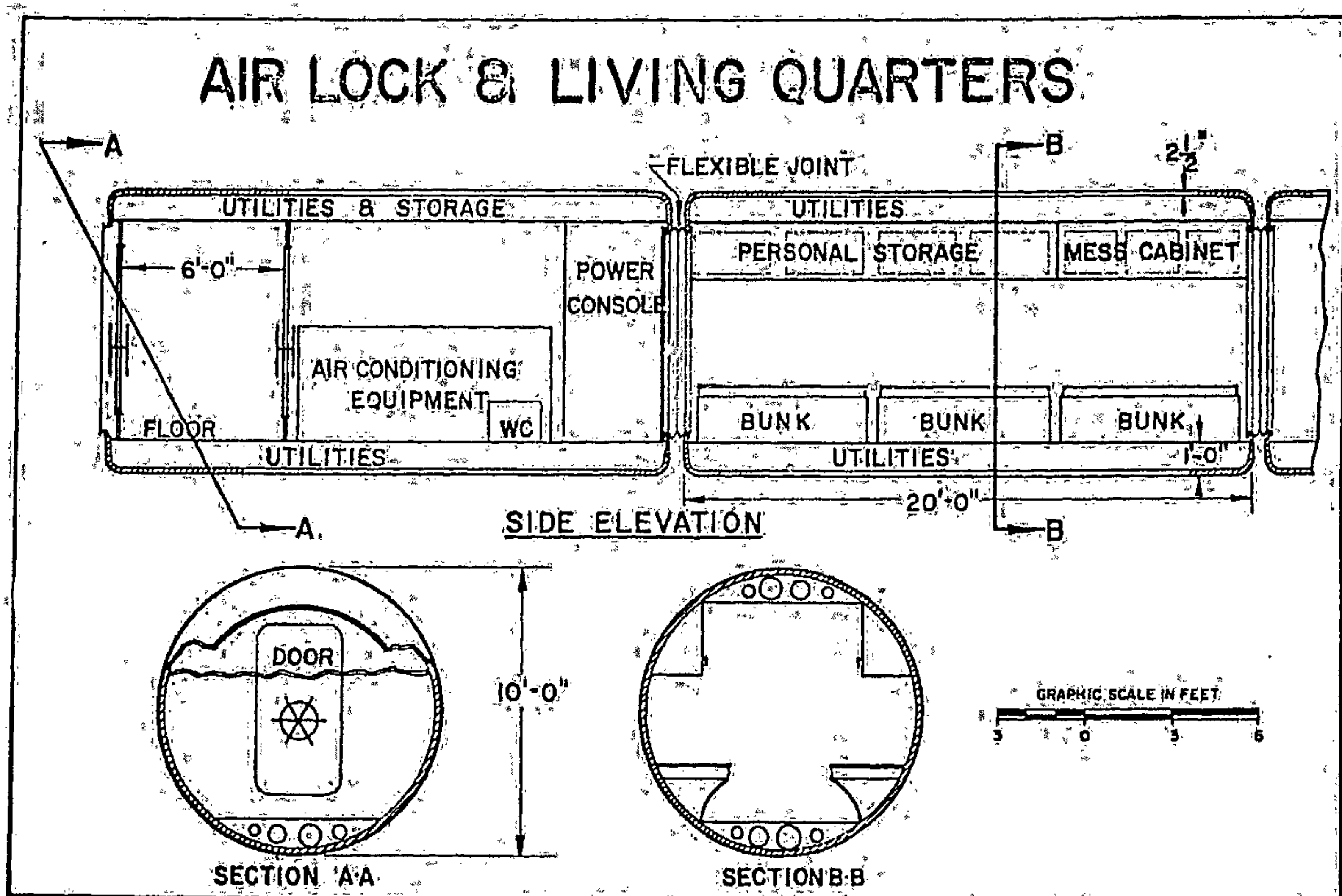


Fig. II-4. Cross-Section of Typical Outpost Compartments

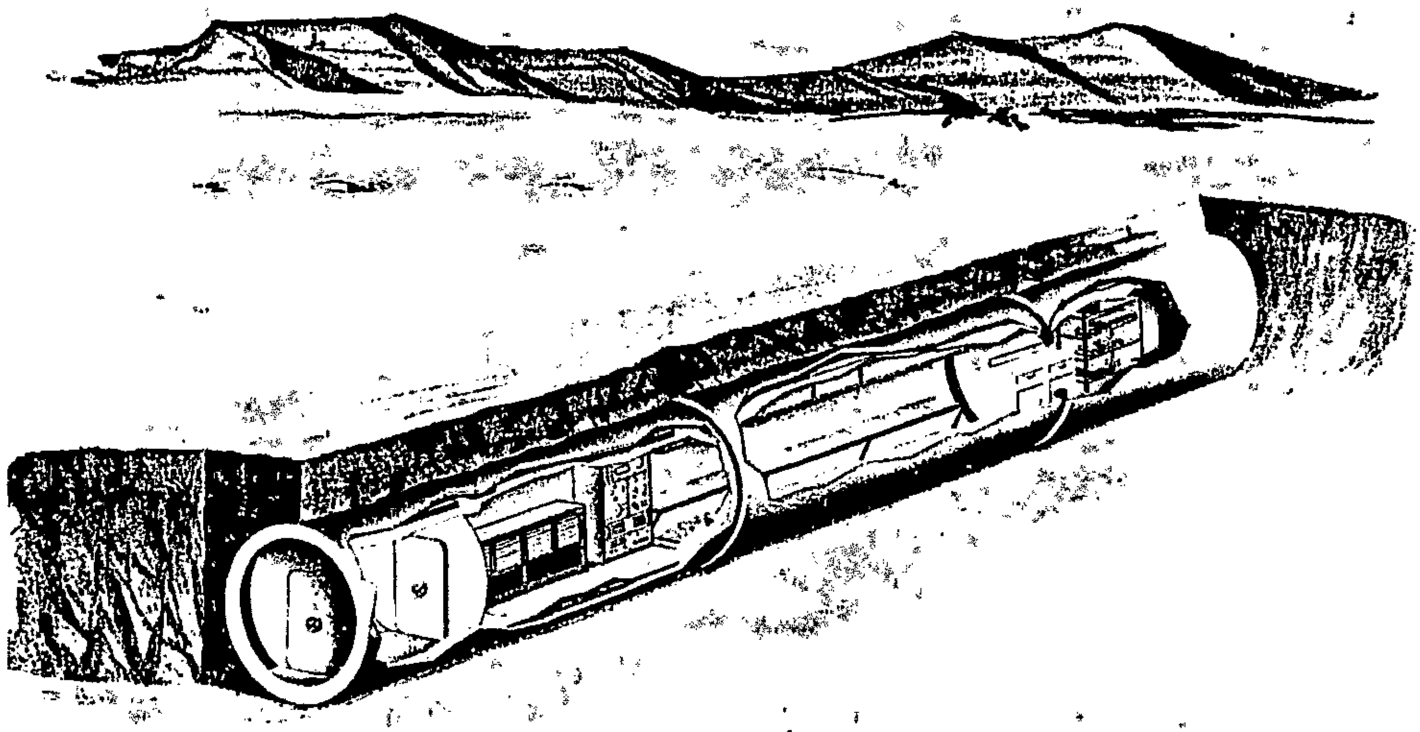


Fig. II-5. Overall View of Initial Construction Camp

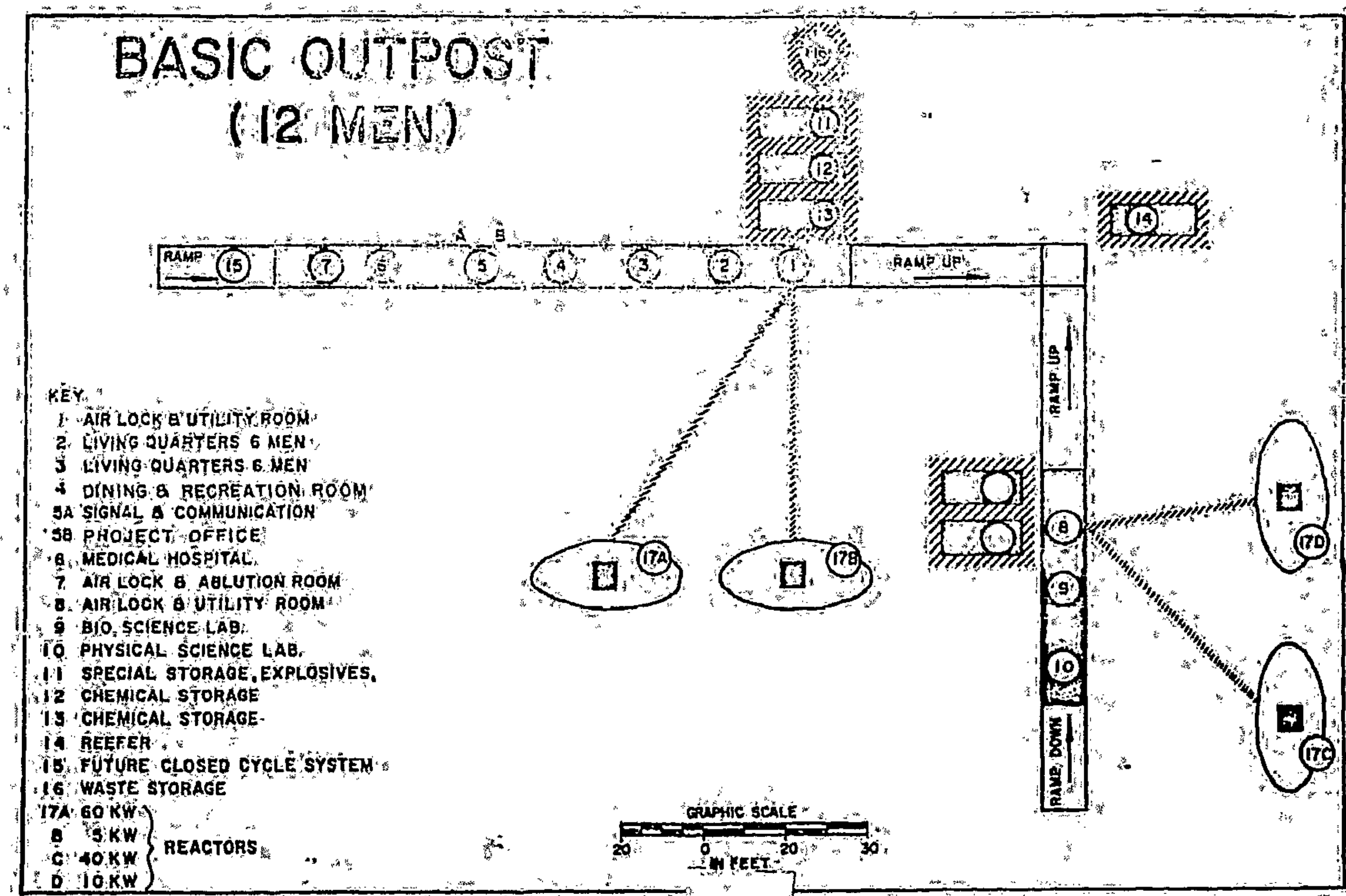


Fig. II-7. Layout Basic 12-Man Outpost

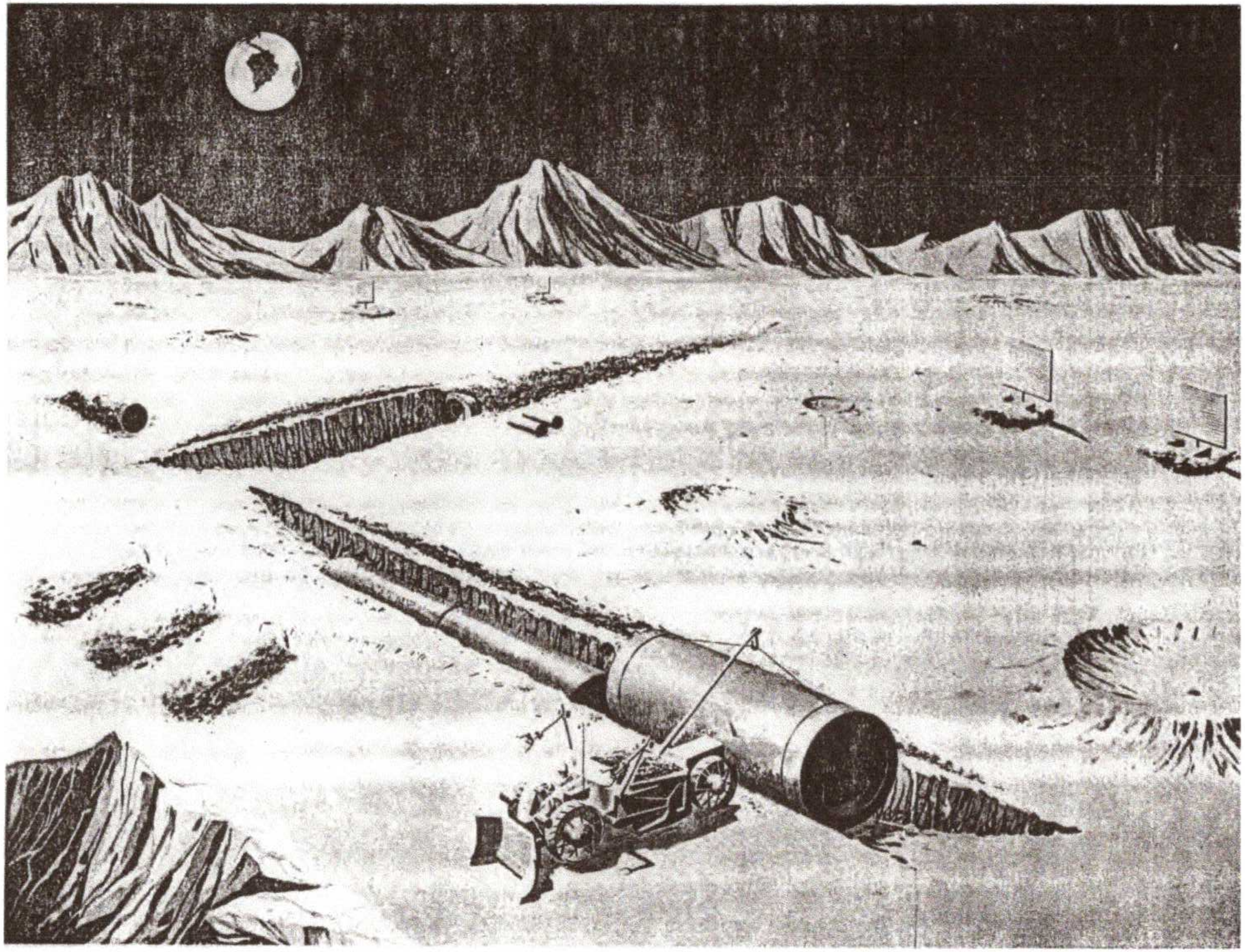


Fig. II-8. Overall View of 12-Man Outpost

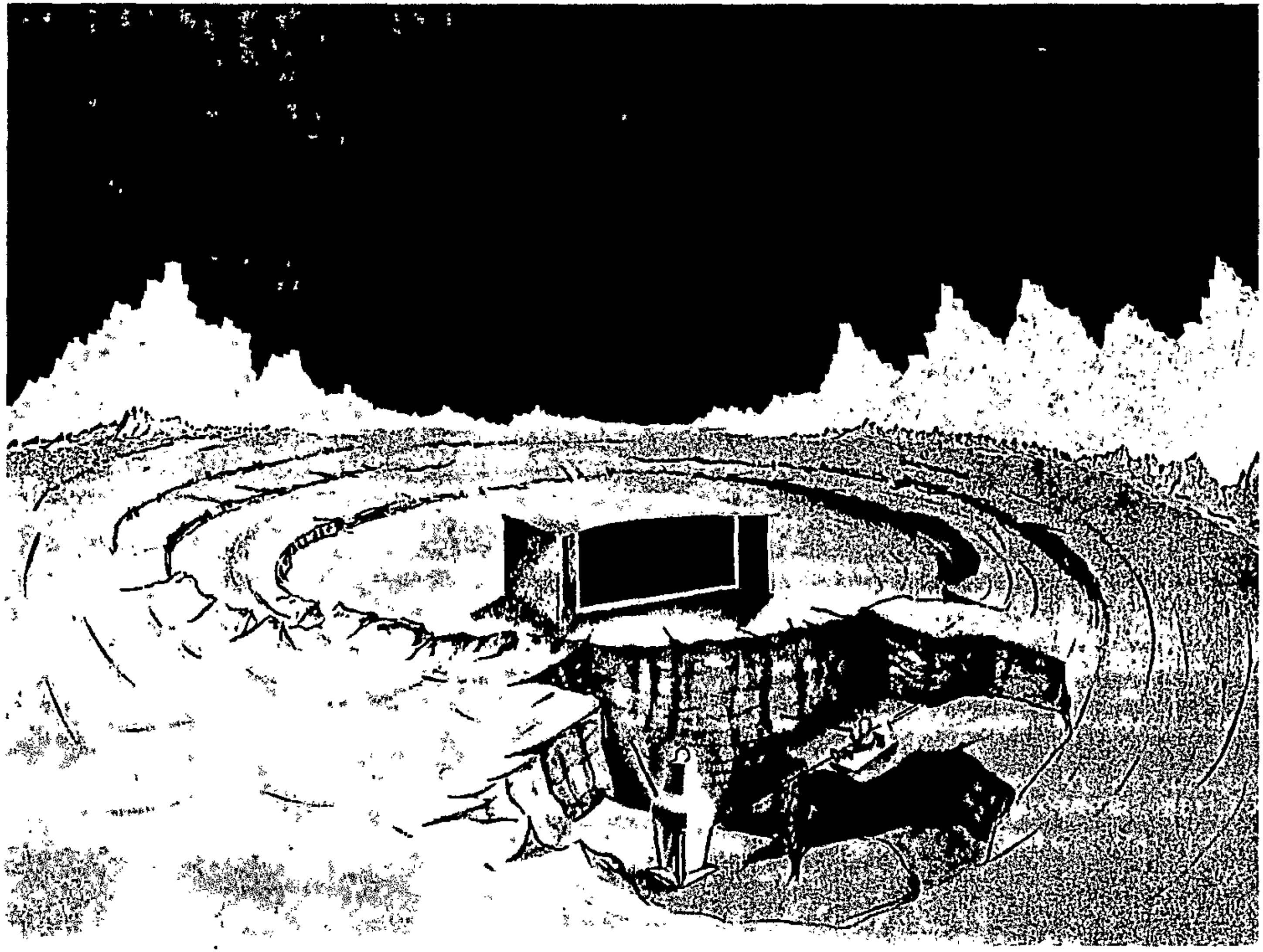


Fig. II-9. Nuclear Power Plant on Moon (60 kw)

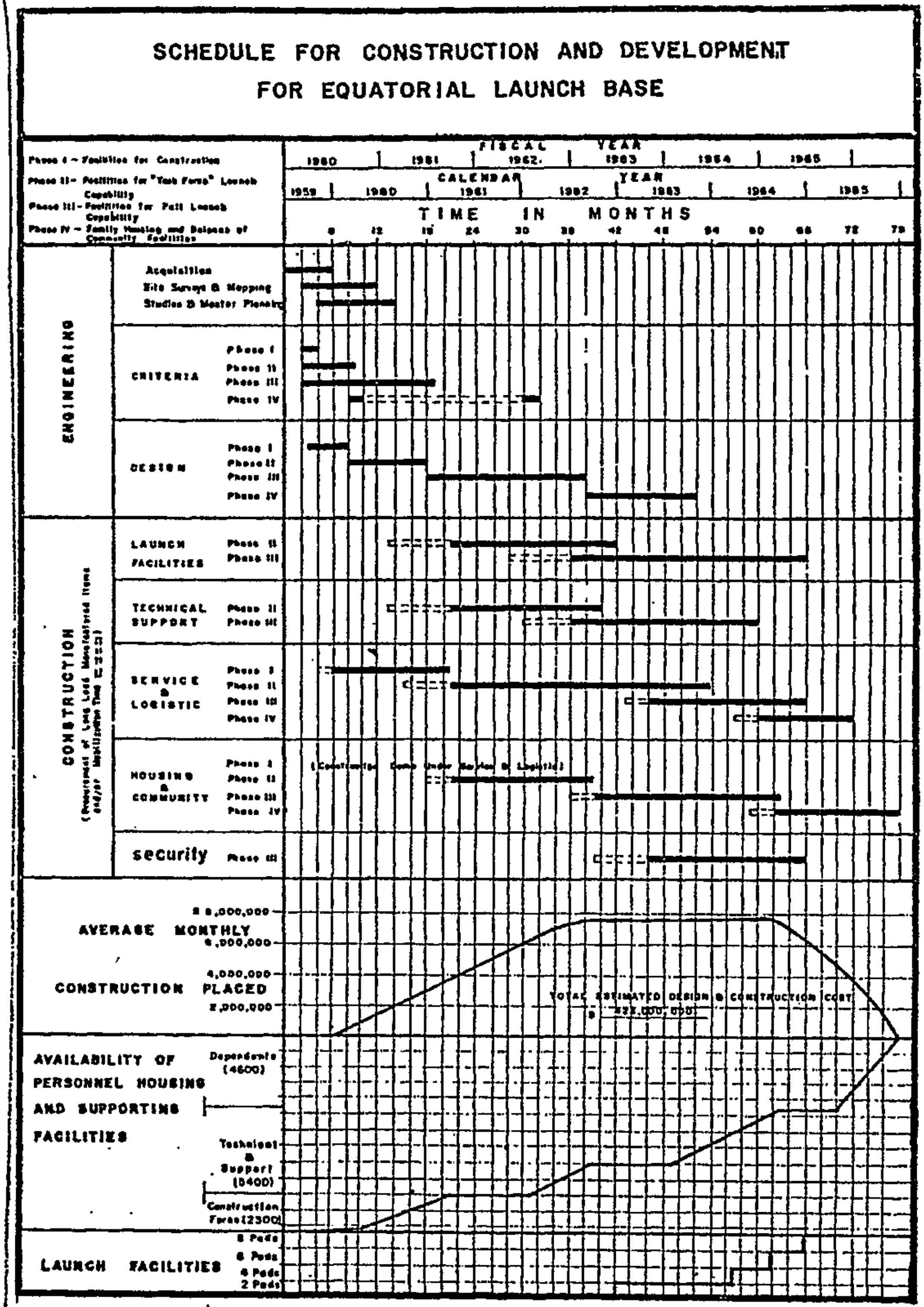
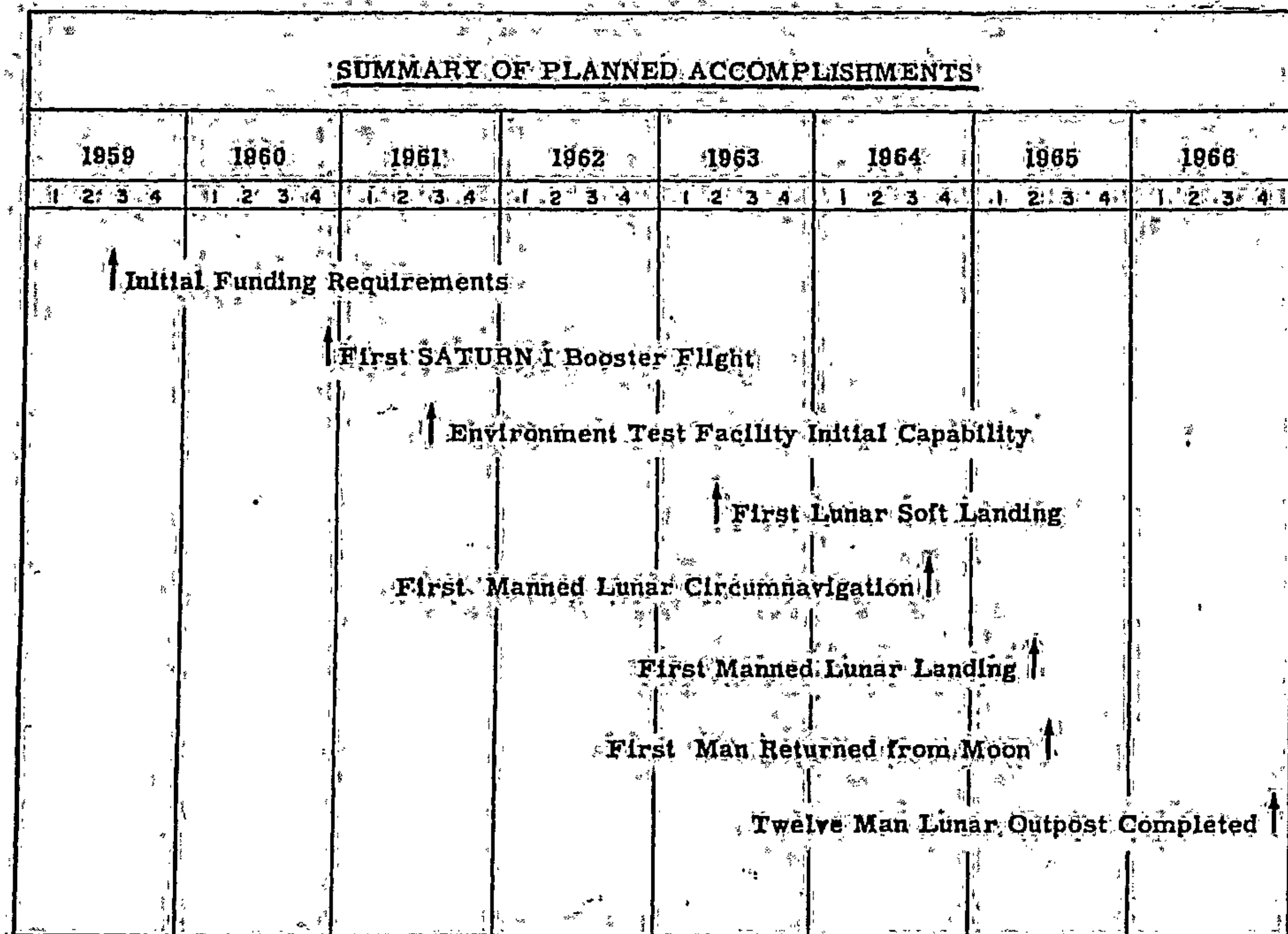


Fig. II-73. Schedule for Construction and Development for Equatorial Launch Base

TABLE II-41

SUMMARY OF HORIZON ACCOMPLISHMENTS



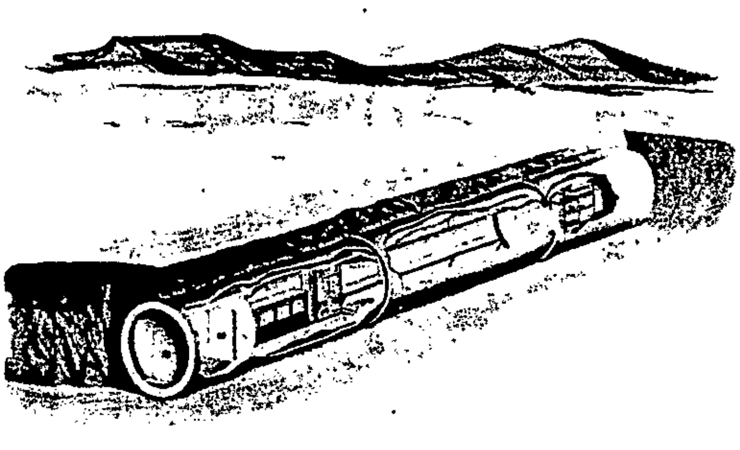


Fig. I-2. Cross Section of Typical Outpost Compartments

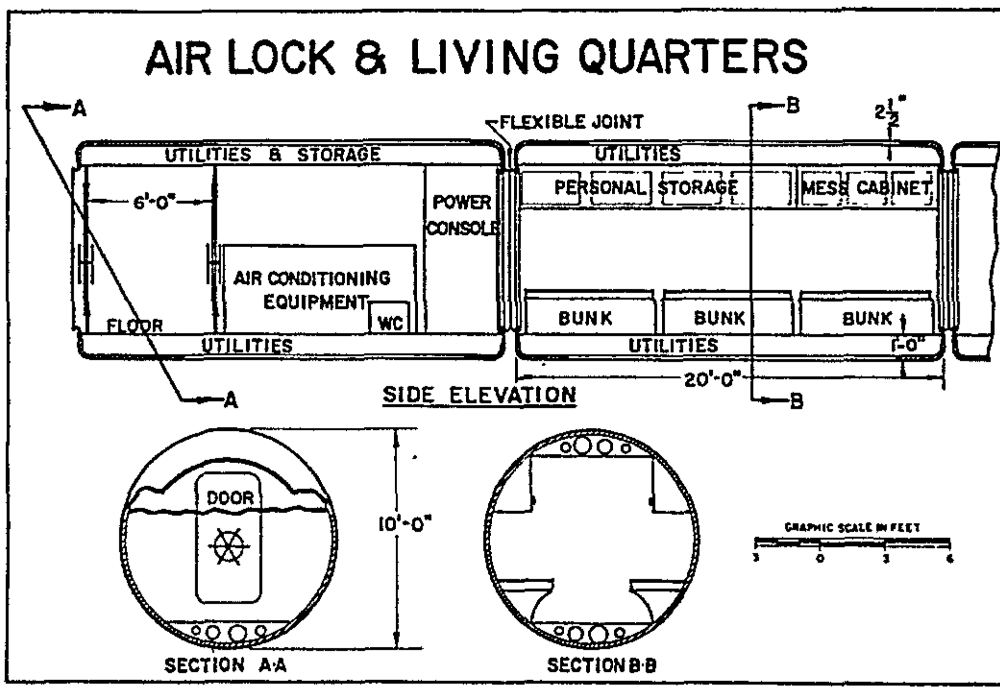


Fig. I-3. Overall View of Initial Construction Camp

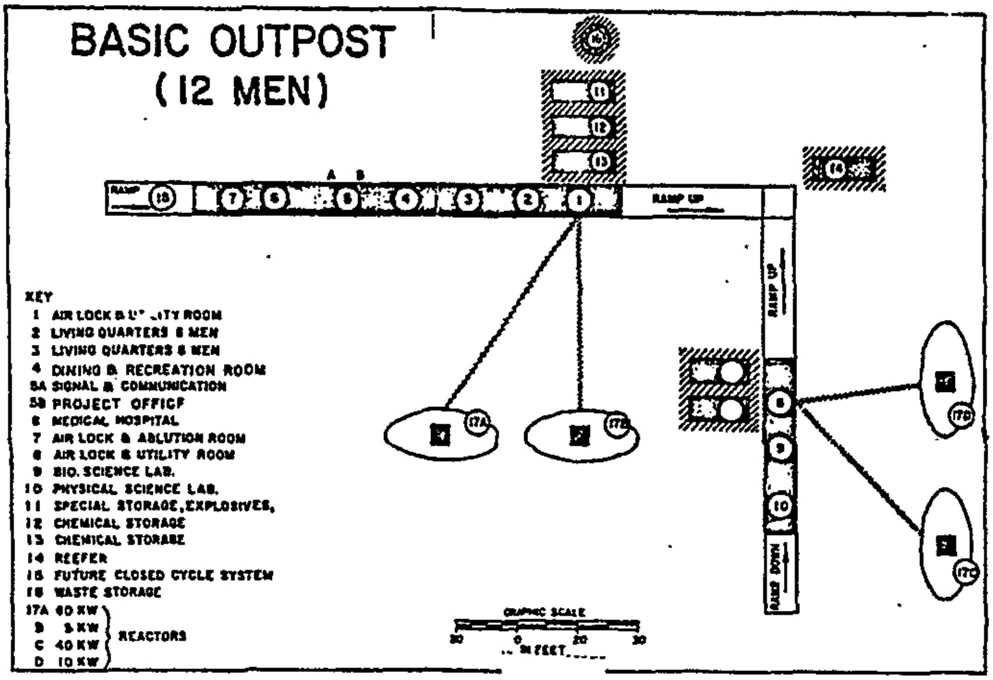


Fig. I-4. Layout of Basic 12-Man Outpost

incidental heat given off by an adequate internal lighting system will nominally supply essentially all of the heat required to maintain comfortable "room" temperature in the outpost quarters.

A suitable atmosphere will be provided within the quarters. The basic gas supply will stem from special insulated tanks containing liquid oxygen or nitrogen. The nitrogen supply needs only to provide for initial pressurization and replacement of leakage losses; whereas, the oxygen is, of course, continuously used to supply bodily needs. However, the weights and volumes of both gases are quite reasonable and presents no unusual problem of supply. Carbon dioxide and moisture will be controlled initially by a solid chemical absorbent and dehumidifier. Such a scheme requires considerable amounts of material; therefore, a carbon dioxide freeze-out system will be installed later.

4. Personnel Equipment

For sustained operation on the lunar surface a body conformation suit having a substantial outer metal surface is considered a necessity for several reasons: (1) uncertainty that fabrics and elastomers can sustain sufficient pressure differential without unacceptable leakage; (2) meteoroid protection; (3) provides a highly reflective surface;

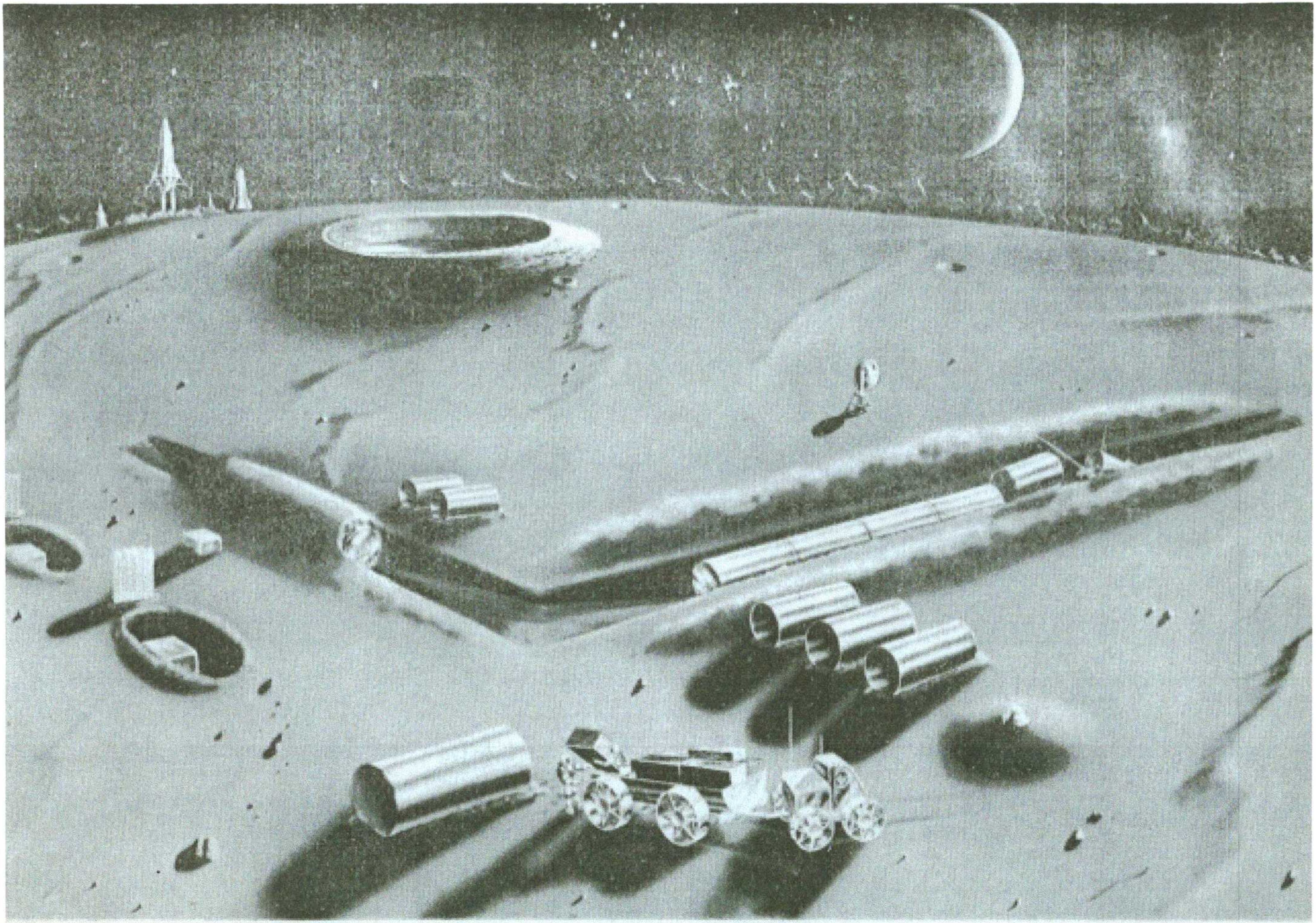


Fig. I-1. HORIZON Outpost in Late 1965